1. Competency #9: Translate a problem into a standard mathematical representation or algorithm

> *Translate a problem into a standard mathematical representation or algorithm*

Highly relevant (⭐⭐⭐⭐⭐)
Students must convert "John's position after 30 minutes" into a **proportional relationship in circular motion**:

- Establish a mapping chain: time → fraction of rotation → angular position → spatial location;
- The core mathematical model is a ratio: $\text{fraction of rotation} = \frac{\text{elapsed time}}{\text{period}} = \frac{30}{40}$.

This is precisely the process of translating a real-world scenario (riding a Ferris wheel) into a **standard mathematical structure (proportion, parametrized circle)**, fully aligning with Competency #9.

---

2. Competency #6: Identify and describe the mathematical elements of a real-world situation, including key variables

> *Identify and describe the mathematical aspects... including identifying the significant variables*

Highly relevant (⭐⭐⭐⭐☆)
Key variables include:

- Total period (40 minutes);
- Elapsed time (30 minutes);
- Starting position (P, lowest point);
- Direction of rotation (indicated by diagram arrow);
- Reference points on the circle (P, R, S).

Students must recognize **uniform circular motion** as the underlying mathematical structure and extract the dependencies among time, angle, and position. Misinterpreting direction or starting point leads to errors, highlighting sensitivity to **key variables and constraints**.

3. Competency #1: Select a mathematical description or representation that captures the problem

> *Select a mathematical description or a representation that describes a problem*

Highly relevant (⭐⭐⭐⭐☆)
The solver must choose an appropriate modeling approach:

- Fractional model: $\frac{30}{40} = \frac{3}{4}$ of a revolution;
- Angular model: $\frac{3}{4} \times 360^\circ = 270^\circ$;
- Parametric unit circle: position = $(\cos\theta, \sin\theta)$ (conceptually implied, though not computed).

Selecting the **fraction-of-revolution** approach is the most efficient, demonstrating understanding of the problem's structure and effective model selection—exactly what Competency #1 emphasizes.

---

4. Competency #8: Recognize connections between the problem and known mathematical concepts or procedures

> *Recognise aspects of a problem that correspond with known problems or mathematical concepts*

Moderately relevant (⭐⭐⭐☆☆)
Students should identify this as a classic **uniform circular motion** or **periodic function** scenario (similar to clock hands or introductory trigonometry on the unit circle). This pattern recognition helps them quickly apply the known principle that **angle is proportional to time**, aligning with Competency #8.

5. Competencies #2, #10, #11: Identify key variables / Use mathematical tools to describe relationships

> *Identify the key variables... Use mathematical tools (symbols, diagrams) to describe relationships*

Moderately relevant (⭐⭐⭐☆☆)
Although only a qualitative answer is required, a full formulation might involve:

- Variables: $t = 30$, $T = 40$;
- Relationship: $\theta = \frac{t}{T} \times 360^\circ$;
- Or using a sketch of the circle to mark the $\frac{3}{4}$ position.

Even if not written down, **mentally constructing this variable relationship** is foundational to modeling, consistent with Competencies #2, #10, and #11.

---

6. Competency #7: Simplify or decompose a situation to make it amenable to mathematical analysis

> *Simplify or decompose a situation... to make it amenable to mathematical analysis*

Moderately relevant (⭐⭐⭐☆☆)
Students must ignore irrelevant details (e.g., riverbed height, exact diameter—neither used for positioning) and focus on:

- Periodicity;
- Starting point;
- Time proportion.

This **abstraction and simplification**—reducing the Ferris wheel to a uniformly rotating circle—is a crucial step in modeling, matching Competency #7.

7. Competency #3: Select a representation appropriate to the problem context

> *Select a representation appropriate to the problem context*

Low relevance (⭐⭐☆☆☆)
While multiple representations (angles, arc length, coordinates) are possible, the problem only requires **qualitative interval judgment**. Most students use a simple "fraction of circle" with a mental or sketched diagram. Thus, representational choice plays a minor role.

---

8. Competencies #4, #5, #12: Decode information / Recognize patterns / Identify assumptions

> *Read/decode tasks... Recognise patterns... Identify constraints/assumptions*

Low to indirectly relevant (⭐☆☆☆☆)

- #4: Requires reading text and diagram (e.g., rotation direction, point labels), but information is clear;
- #5: Recognizing "constant speed" and "periodicity" as patterns is straightforward;
- #12: Implicit assumptions include "uniform speed," "no stops," and "P is the lowest point," but the problem doesn't ask students to reflect on them.

These competencies are present but not central to the modeling challenge.

1. 第9条：将问题转化为标准数学表示或算法

> *Translate a problem into a standard mathematical representation or algorithm*

高度相关（⭐⭐⭐⭐⭐）
学生需将“30分钟后的物理位置”转化为**圆周运动中的角度或弧长比例**：

- 建立 时间 ↔ 圈数 ↔ 角度 ↔ 位置 的映射链；
- 核心数学模型是 比例关系： $\text{fraction of rotation} = \frac{\text{elapsed time}}{\text{period}} = \frac{30}{40}$ 。
  这正是将现实情境（坐摩天轮）翻译为标准数学结构（比例、圆周参数化） 的过程，完全契合第9条。

---

2. 第6条：识别并描述现实问题中的数学要素，包括关键变量

> *Identify and describe the mathematical aspects... including identifying the significant variables*

高度相关（⭐⭐⭐⭐☆）
关键变量包括：

- 总周期（40 分钟）；
- 经过时间（30 分钟）；
- 起始位置（P，最低点）；
- 旋转方向（由图示箭头确定）；
- 圆周上的关键参考点（P, R, S）。

学生必须识别“**匀速圆周运动**”这一核心数学结构，并提取时间、角度、位置之间的依赖关系。忽略旋转方向或起始点都会导致错误，凸显了对**关键变量和约束**的敏感性。

---

3. 第1条：选择能描述问题的数学表述或模型

> *Select a mathematical description or a representation that describes a problem*

高度相关（⭐⭐⭐⭐☆）
解题者需主动选择合适的建模视角：

- 比例模型：30/40 = 3/4 圈；
- 角度模型：3/4 × 360° = 270°；
- 单位圆参数化：位置 = $(\cos\theta, \sin\theta)$ （虽不需计算，但概念隐含）。

选择“**圈数比例**”是最直接有效的策略，体现了对问题结构的理解和模型选择能力，符合第1条。

4. 第8条：识别问题中与已知数学概念或程序的对应部分

> *Recognise aspects of a problem that correspond with known problems or mathematical concepts*

中等相关（⭐⭐⭐☆☆）
学生需识别这是**匀速圆周运动**或**周期函数**的典型应用（类似钟表指针、单位圆三角函数入门问题）。这种“模式识别”帮助调用已有知识：**时间与角度成正比**，从而快速建立比例模型，符合第8条。

5. 第2条 & 第10/11条：识别关键变量 / 使用数学工具描述关系

> *Identify key variables... Use mathematical tools (symbols, diagrams) to describe relationships*

中等相关（⭐⭐⭐☆☆）
虽然最终只需定性判断（在哪个区间），但完整建模可引入：

- 变量 $t = 30$, $T = 40$;
- 关系式 $\theta = \frac{t}{T} \times 360^\circ$;
- 或借助圆周示意图标记四分之三位置。

即使未写出公式，**在脑中构建变量关系**仍是建模基础，符合第2、10、11条。

---

6. 第7条：简化或分解问题以利于数学分析

> *Simplify or decompose a situation... to make it amenable to mathematical analysis*

中等相关（⭐⭐⭐☆☆）
学生需忽略无关细节（如河床高度、直径具体数值——本题未用于定位），聚焦于：

- 周期性；
- 起始点；
- 时间比例。

这种**抽象与简化**（将摩天轮简化为一个匀速转动的圆）是建模的关键一步，符合第7条。

---

7. 第3条：选择适合问题情境的表示方式

> *Select a representation appropriate to the problem context*

低相关（⭐⭐☆☆☆）
虽然可用角度、弧长、坐标等多种表示，但本题只需**定性区间判断**，因此通常直接用“圈数分数”配合圆周草图即可。表示方式选择空间有限，故相关性较低。

8. 第4、5、12条：解码信息 / 识别结构 / 识别假设

> *Read/decode tasks... Recognise patterns... Identify constraints/assumptions*

低至间接相关（⭐☆☆☆☆）

- 第4条：需读懂文字和图示（如旋转方向、点标签），但信息清晰；
- 第5条：识别“匀速”和“周期性”是规律，但较直接；
- 第12条：隐含假设包括“匀速”“无停顿”“P 为最低点”，但题目未要求反思这些。

这些能力存在，但非本题建模的主要挑战。